COURSE PLAN

Department	MCA
Course Title	Advanced Programming using Python
Course Code	25CSA632
Programme and Section	MCA ‘C’ & ‘D’
Academic Year	2026-27
Semester	III
LTP/ hours per week- Credit	L: 03, T: 00 and P: 01, 4 Credits
Course Instructor Details	Dr. Rachaita Dutta rachaitadutta@my.amrita.edu +91 90072 23301 Ms. Priyanka M priyankam@my.amrita.edu +91 9060486121

Programme Outcomes:

PO1	**Computational Knowledge:** Apply knowledge of computing fundamentals, computing specialization, mathematics, and domain knowledge
PO2	**Problem Analysis:** Identity, formulate, research literature, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
PO3	**Problem Analysis:** Identity, formulate, research literature, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
PO4	**Design / Development of Solutions:** Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.
PO5	**Conduct Investigations of Complex Computing Problems:** Use research-based knowledge and research methods, including design of experiments, analysis, and interpretation of data, and synthesis of the information to provide valid conclusions

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PO6	**Modern Tool Usage:** Create, select, adapt and apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations.
PO7	**Professional Ethics:** Understand and commit to professional ethics and cyber regulations, responsibilities, and norms of professional computing practice.
PO8	**Life-long Learning:** Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional.
PO9	**Communication Efficacy:** Communicate effectively with the computing community, and with society at large, about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions.
PO10	**Societal and Environmental Concern:** Understand and assess societal, environmental, health, safety, legal, and cultural issues within local and global contexts and the consequential responsibilities relevant to professional computing practice.
PO11	**Individual and Teamwork:** Function effectively as an individual and as a member or leader in diverse teams and in multidisciplinary environments.
PO12	**Innovation and Entrepreneurship:** Identify a timely opportunity and using innovation to pursue that opportunity to create value and wealth for the betterment of the individual and society at large.

Course Outcomes (COs):

CO1	Apply core and object-oriented programming principles in Python.
CO2	Use advanced Python features such as iterators, generators, decorators, and context managers.
CO3	Develop robust Python applications using modules, error handling, and unit testing.
CO4	Implement algorithmic problem-solving techniques using Python.
CO5	Handle text data using regular expressions and work with file systems efficiently.

Prerequisites:

- Basic Programming Concepts and Fundamentals of Python
- Problem Solving and Basic Data Structures

Lesson Plan

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Week	Lecture No.	Topics	Keywords	Teaching Aids	Textbook with chapter	CO Mapping
1	01	**Unit 1: Introduction & Object-Oriented Programming**	Python Features, Applications, Installation, IDEs (VS Code), Python Interpreter, Interactive Mode, Script Mode, Comments, Indentation	Presentation & demonstration	TB1, CH1	1
	02	Python Syntax & Data Types	Variables, Naming Conventions, Keywords, Literals, Numeric Types, Strings, Boolean, Type Conversion, Input/Output	Blackboard & demonstration	TB1, CH2	1
	03	Control Flow	Conditional Statements (if, elif, else), Loops (for, while), Nested Loops, break, continue, pass, range()	Blackboard, Presentation & demonstration	TB1, CH3	1
2	04					
	05	Functions	Function Definition, Function Calling, Parameters, Default Arguments, Keyword Arguments, Variable-Length Arguments (*args, **kwargs), Scope, Return Statement	Blackboard, Presentation & demonstration	TB1, CH5	1
	06					
3	07	Modules & Packages	Import Statements, Creating Modules, Packages, Built-in Modules, User-defined Modules	Presentation & demonstration	TB1, CH6; TB2, CH5	3
	08	Lambda and Recursion Functions	Anonymous Functions, lambda, map(), filter(), reduce(); Recursive Functions, Recursion Tree, Tail Recursion	Blackboard, Presentation & demonstration	TB1, CH5; TB2, CH7	1
	09	File Handling	Opening Files, File Modes, Reading/ Writing Text Files, File Objects, with Statement, File Pointer Operations		TB1, CH7; TB2, CH8	3
Class Test 01						
4	10	Exception Handling	Exception Types, try, except, finally, raise, assertions and user-defined exceptions	Blackboard, Presentation & demonstration	TB1, CH6; TB2, CH5	3
	11	Debugging techniques	Assertions, Exception Handling, Logging, Python Debugger, Breakpoints & Call Stack			3

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	12	Object-Oriented Programming	Classes, Objects, Attributes, Methods, Instance Variables, Constructors, self, Object Lifecycle	Blackboard, Presentation & demonstration	TB1, CH12; TB2, CH1	1
5	13	Advanced OOP	Inheritance & its types, Polymorphism, Operator Overloading, Method Overriding, Composition vs Inheritance, Class Methods, Static Methods			1
	14					
	15	Magic Methods	__init__, __str__, __repr__, __len__, __add__, __eq__	Presentation & demonstration	TB2, CH1	1
6	16	**Unit 2: Advanced Python Programming Concepts:** Iterators	Iterable Objects, Iterator Protocol, __iter__(), __next__(), Custom Iterators	Blackboard, Presentation & demonstration	TB2, CH17	2
	17	Generators	Generator Functions, yield, Generator Expressions, Lazy Evaluation, Memory Efficiency			2
	18	Decorators	Function Decorators, Nested Functions, Wrappers, Decorator Syntax, Multiple Decorators	Blackboard, Presentation & demonstration	TB2, CH9	2
7	19	Closures	Nested Functions, Lexical Scope, Free Variables, Closure Objects, Practical Examples			2
	20	Context Managers	Resource Management, with Statement, __enter__(), __exit__(), Contextlib Module	Blackboard, Presentation & demonstration	TB2, CH18	2
	21	Modules and packages	Standard (collections, functools, itertools, math, datetime, random, sys) and create and import custom modules; and package (create, import, _init_)		TB1, CH6; TB2, CH6	2
Midterm Examination						
8	22	Dynamic Typing	Duck Typing, Type Checking, Mutable vs Immutable Objects, Object References	Blackboard, Presentation & demonstration	TB2, CH2	2
	23	Introspection	type(), id(), dir(), getattr(), setattr(), hasattr(), Reflection			
	24	**Unit 3: File and Text Processing:** Text File	Reading Large Files, Line-by-Line Processing, Encoding, File Buffering	Presentation & demonstration	TB1, CH7; TB2, CH8	3

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9	25	Binary Files	Binary Read/Write, Serialization, Pickle Module, Byte Streams	Presentation & demonstration	TB2, CH8	3
	26	Directory Handling	os, os.path, pathlib, shutil, glob, Directory Traversal	Presentation & demonstration	TB2, CH8	3
	27	JSON and CSV Processing	JSON Structure, Serialization, Deserialization, json Module; CSV Reader, CSV Writer, DictReader, DictWriter, Data Cleaning	Presentation & demonstration	TB2, CH6	3
10	28	Regular Expressions using the re module	search, match, replace, extract patterns		TB2, CH4	5
	29	Usecase: Log file analyzer	Log Parsing, Timestamp Extraction, Pattern Matching, Error Detection, Report Generation	Presentation & demonstration	TB2, CH4; TB3, CH2	5
	30	**Unit 4: Problem-Solving Patterns in Python:** Sorting	Bubble Sort, Selection Sort, Insertion Sort, Merge Sort, Quick Sort, Python sorted()	Blackboard, Presentation & demonstration	TB1, CH13	4
11	31	Searching	Linear Search, Binary Search, Search Complexity, Practical Applications			4
	32	Recursion	Recursive Thinking, Base Case, Recursive Case, Stack Memory, Memorization	Blackboard, Presentation & demonstration	TB1, CH5	4
	33	Hashing & Frequency Counter	Dictionary, Hash Tables, Counting Techniques, Character Frequency, Word Frequency		TB1, CH13	
	34	Sliding Window Technique	Fixed Window, Variable Window, Maximum Sum Subarray, Longest Substring Problems	Blackboard, Presentation & demonstration		4
12	35	Backtracking	State Space Tree, Constraint Satisfaction, N-Queens, Sudoku Solver, Rat in a Maze			
	36	Basic graph and tree problems using dictionaries/lists	Graph Representation, Adjacency List, BFS, DFS, Connected Components; Binary Tree Representation, Tree Traversals, Recursion on Trees, Dictionary/List Representation; Time Complexity, Space Complexity, Coding Patterns, LeetCode Problems, HackerRank Challenges	Blackboard, Presentation & demonstration	TB1, CH13	4
	37	Competitive Programming				

	38	**Unit 5: Testing and Best Practices:** Writing Testable Code	Test Cases, Assertions, Test Design Principles, Modular Programming	Presentation & demonstration	TB3, CH1	4
	39	Unit Testing	Unit Test Framework, Test Suites, Fixtures (setUp(), tearDown()), Mock Objects, Test Automation		TB3, CH2–CH4	
14	40	PyTest Framework	Test Discovery, Fixtures, Parameterized Tests, Markers, Plugins, Code Coverage, Test execution using pytest	Presentation & demonstration	TB3, CH5	3
	41	Code Documentation and style (PEP8)	Comments, Docstrings (PEP 257), API Documentation, Sphinx Basics, Type Hints; PEP 8 Style Guide, Naming Conventions, Coding Standards (Code Formatting, Indentation, Imports, Line Length, Readability, Static Analysis Tools (flake8, pylint))	Presentation & demonstration	TB2, CH15; TB3, CH6	3
	42	Error Logging and Debugging	Logging Levels (DEBUG, INFO, WARNING, ERROR, CRITICAL), Log Formatting, Exception Logging, Debugging: Tracebacks, pdb, Breakpoints;	Presentation & demonstration	TB3, CH6; TB2, CH15	5
15	43	Packaging Python applications	Virtual Environments (venv), pip, Package Installation, Requirements File (.txt), Package & Project Structure, Packaging Applications, Requirement Analysis, Version Control, Deployment Basics	Presentation & demonstration	TB2, CH15; TB3, CH6	5
	44	Git and GitHub basics	Git Workflow, Repository, Branching, Commit, Push, Pull, Merge, GitHub Collaboration	Presentation & demonstration	TB3, CH6	5
	45	Clean Code Practices	Code Refactoring, Naming Conventions, Modular Design, Design Principles, Code Smells			5
End Semester Examination						

Lab Lesson Plan

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Week	Topic	Teaching Aid	Textbook with chapter	CO Mapping
1	Python installation, IDE setup (VS Code/PyCharm), Python Shell, Variables, Data Types, Operators, Input/Output, Conditional Statements, Loops, Pattern Programs	Demonstration	TB1, CH1–CH3	1
2	User-defined Functions, Function Arguments, Default Arguments, Variable-length Arguments (*args, **kwargs), Lambda Functions, Recursive Functions, Creating Modules, Importing Modules	Demonstration	TB1, CH5–CH6	1
3	Classes, Objects, Constructors, Instance Variables, Methods, Encapsulation, Method Overriding, Polymorphism, Class Methods, Static Methods	Demonstration	TB1, CH12; TB2, CH1	1
4	Magic Methods (__init__, __str__, __len__, __repr__), Operator Overloading, Composition vs Inheritance, Mini Case Study (Student/Employee Management System)	Demonstration	TB2, CH1	1
5	Custom Iterators, Iterator Protocol, Generator Functions, yield, Generator Expressions, Decorators, Function Wrappers, Closures	Demonstration	TB2, CH9; CH17	2
6	Context Managers (with Statement), Creating Custom Context Managers, Dynamic Typing, Object Introspection (dir(), type(), id(), getattr(), setattr()), Standard Modules (collections, functools)	Demonstration	TB2, CH18; CH2	2
7	Text Files, Binary Files, File Modes, Reading/Writing Files, Exception Handling, Directory Operations (os, pathlib, shutil, glob)	Demonstration	TB1, CH7; TB2, CH8	3
8	JSON Serialization & Deserialization, CSV Reader/Writer, Data Extraction, Data Cleaning, Exception Handling during File Processing	Demonstration	TB2, CH6; CH8	3
9	Pattern Matching, Character Classes, Quantifiers, Groups, search(), match(), findall(), sub(), Email Validation, Phone Validation, Password Validation, Log File Analysis	Demonstration	TB2, CH4; TB3, CH2	
10	**Lab Test 01**			
11	Sorting Algorithms, Searching Algorithms, Dictionary Operations, Hashing, Frequency Counter Problems, String Manipulation	Demonstration	TB1, CH13	5
12	Sliding Window Technique, Recursion Problems, Backtracking (N-Queens, Rat in a Maze), Time Complexity Analysis	Demonstration		4

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13	Graph Representation using Dictionaries, BFS, DFS, Binary Tree Representation, Tree Traversals, Competitive Programming Problems	Demonstration		4
14	Writing Test Cases using unittest, Test Suites, Assertions, Debugging using pdb, Logging Module, Exception Logging	Demonstration	TB3, CH2–CH6	4
15	PEP 8 Coding Standards, Code Refactoring, Docstrings, Type Hints, Packaging Python Applications, Virtual Environments, Git & GitHub Repository Management	Demonstration	TB2, CH15; TB3, CH6	3

CO-PO Mapping

PO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	2	3	-	-	-	1	-
CO2	3	3	3	1	-	-	1	-
CO3	2	3	3	2	1	-	2	-
CO4	3	3	3	2	1	-	2	-
CO5	1	1	1	2	2	-	1	-

Session by Alumni/ Industry Expert:

1. Advanced Python in Software Development and for Automation – 10^{th} Week

Textbooks

1. Python Programming: An Introduction to Computer Science by John Zelle
2. Fluent Python by Luciano Ramalho
3. Test-Driven Development with Python by Harry J.W. Percival

Internal Assessment Components

Sl. No.	Component Name	Weightage	CO Mapping	Tentative Date of Completion
1	Class Test 01	05	1, 2	4th week
2	Midterm Examination	20	1, 2	8th week
3	Lab Test 01	20	1, 2, 3, 4	10th Week
4	Continuous Assessment	10	1, 2, 3, 4, 5	End of the semester
5	Team Project (PORTFOLIO)	10	1, 2, 3, 4, 5	End of the semester
6	Class Participation/ NPTEL CERTIFICATION	05	1, 2, 3, 4, 5	End of the semester

Signatures with name:

Faculty: Dr. Rachaita Dutta 12/8/26 Ms. Priyanka M 12/08/26

Vivek Chinmay.R LIKHITHA.A

Student Representative:

SANJAY.S (MY.Ac.P2MCA25240)

Rakshitha.T C (MY.Ac.P2MCA25112)

Class Advisor Program Coordinator 22/8/26 Chairperson 22-Aug-26

Dr. Rachaita Dutta